Modern Statistical Methods

Homework 3

April 6, 2025

This homework involves software coding. You may choose any software preferred. Each subquestion is worth 10 points. To receive full credit, you need to **(1)** print out your complete code **(2)** write up the necessary information and explanation that leads to the code **(3)** report the results appropriately. It is the course policy that any plagiarism will lead to zero credit or more severe consequence.

1. Generate your data as follows: let $(X_1, \epsilon_1, Y_1), \ldots, (X_{300}, \epsilon_{300}, Y_{300})$ be i.i.d. random vectors with each $X_i \sim N(0,1)$, $\epsilon_i \sim N(0, 0.5^2)$ and independent from X_i, and $Y_i = X_i + \sqrt{|X_i|}\epsilon_i$. Let $\widehat{\beta}$ be OLS for the linear regression model derived from this dataset.

   (a) Use the naive bootstrap method to estimate the variance of $\widehat{\beta}$.

   (b) Use the wild bootstrap method to estimate the variance of $\widehat{\beta}$, where W_i is generated as $P(W_i = -1) = P(W_i = 1) = .5$.

   (c) Comment on the results from the two methods above.

2. Select a statistical method and a specific estimator therein that you are familiar with and would like to investigate (for example, the lasso estimator of a nonzero coefficient in linear regression). Generate your own data appropriately. Denote this estimator by $\widehat{\theta}$, and denote the corresponding parameter by θ_0.

   (d) Use the bootstrap method to estimate the bias of $\widehat{\theta}$, and explain how large B should be theoretically.

   (e) Find a 95% confidence interval for θ_0 using the percentile method.

   (f) Find a 95% confidence interval for θ_0 using the percentile-t method.

   (g) Comment on the results from (e) and (f).

3. Suppose we have an i. i. d. sample of size $n = 1000$ generated from the uniform distribution on $(0, 1)$.

   (h) Use the bootstrap method with $B = n$ to simulate the distribution of the 75% sample quantile (show the histogram).

   (i) Discuss how large B should be in order to give a reliable estimate of the variance of the sample quantile described above.

   (j) Suppose now we are interested in a more extreme sample quantile, that is, the 99.5% sample quantile in this dataset. Does the bootstrap method still give you the same type of simulated distribution as above? If not, discuss the underlying reason in theory, or illustrate how enlarging n may address the issue.